

HARISH MADURAIMANI S

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Profile

Electronics Engineer with strong interdisciplinary expertise in Artificial Intelligence, Embedded Systems and Data Science. Skilled in developing intelligent systems combining hardware and software, including IoT solutions, machine learning models and real-time applications. Passionate about building scalable AI driven products and solving real world problems through innovation in signal processing, analog digital electronics and generative AI. Currently expanding expertise in PCB Design, RTL Design, and GDS flow using Verilog, along with system level design and hardware software co design.

Projects

1. AI SINGLE COPILOT SYSTEM

- Developed an AI-based decision support system for aviation assistance
- Integrated real-time data such as fuel, weather and aircraft parameters
- Implemented predictive models for alternate route and fuel optimization

2. SMART CAR PARKING SYSTEM

- Designed an IoT based parking management system
- Used sensors to detect slot availability in real time
- Improved parking efficiency and reduced congestion

3. Local Business Prediction AI System

- Developed AI chatbot to suggest profitable business ideas based on location
- Includes risk analysis, ROI prediction and demand mapping
- Integrated Google Maps and trend analysis concepts

4. Smart Dustbin (IoT-Based)

- Designed an automated waste management system

5. SafeNest – Disaster-Resilient IoT Communication System

- Built SafeNest, an ESP32 and LoRa based emergency communication system.
- Enabled long range transmission of GPS and biological data without cellular or internet connectivity.
- Implemented reliable wireless communication using ACK based LoRa protocol.

6. Mining Rescue Rover – Disaster-Resilient IoT Communication System

- Autonomous mining rover for exploration and monitoring in hazardous mining environments.
- Integrated robotics, sensors, navigation and wireless communication for remote operation and real-time data collection.

Education-Bachelor's Degree in Electronics Engineering

RAJALAKSHMI INSTITUTE OF TECHNOLOGY, CHENNAI

Skills & Abilities

Embedded Systems

- ESP32, ESP32-S3, STM32, 8051, Raspberry Pi 5, ARM Cortex-M, Embedded C, Microcontrollers, Embedded Linux, FreeRTOS, Sensors & Actuators

IoT & Wireless

- LoRa, Wi-Fi, Bluetooth, GPS, GSM, MQTT, Blynk IoT, ThingSpeak, Adafruit IO, Node-RED, HiveMQ, ThingsBoard

Electronics

- Analog & Digital Electronics, Circuit Design, LTspice, PCB Design, KiCad, Altium, VLSI, Verilog, Microwind

Robotics

- ROS2, Gazebo, MATLAB, Simulink, Automation, Control Systems, Scilab, Fusion 360, Blender

Programming

- Python, C, C++, Rust, CUDA, Java, SQL, Data Structures & Algorithms

Web Development

- HTML, CSS, JavaScript, React.js, Django, Node.js, Firebase, MongoDB, Authentication, Payment Gateways

AI / ML

- Machine Learning, Deep Learning, Computer Vision, YOLO, U-Net, OpenCV, PyTorch, TensorFlow, NLP, CNN, RNN, LLMs, VLMs, Edge AI

Generative AI

- LangChain, LCEL, LangGraph, RAG Systems, Vector Databases, pgVector, LangSmith

DevOps & Tools

- Git, GitHub, Linux, Shell Scripting, Vercel, Netlify, Postman, Playwright

Design & Analytics

- UI/UX, Figma, Power BI, Google Analytics, Google Search Console